SRN: 2 5 S O O B F A D S / 6

Internal Assessment – 1 (CIA-1)

Department:	Computer Science & Engineering	**Date:**	11/12/2025
Course Name:	Problem Solving Using C	**Course Code:**	25BEPHY104/25BEEL
Batch:	2025-26	**Max. Marks:**	50
Semester:	I	**Duration:**	90 Mins

Note: Answer all the questions. Q1 & Q2 carries 10 Marks each, Q3 - Q6 carry 20 Marks eac

Q No	Question	Marks	RBT Level
1.	a. What is a variable? Explain different rules to define a variable name with examples.	5	L2
	b. Differentiate between variables and identifier.	5	L2
	OR		
2.	a. What is decision control? Illustrate its importance in programming.	5	L3
	b. Differentiate between sequential, decision-making, and looping constructs.	5	L3
3.	a. What are format specifiers? List any four with their meanings.	6	L3
	b. Describe different types of constants in C with suitable examples.	6	L2
	c. What are identifiers? State the rules for naming them.	8	L3
	OR		
4.	a. Explain the use of **Scanf()** with an example.	6	L2
	b. Discuss the importance of header files in C programming.	6	L2
	c. List different characteristics of C programming	8	L2
5.	a. Write a program to calculate the factorial of a number using a for loop.	10	L3
	b. Write a program to find the sum of digits of a number using a **while loop**.	10	L3
	OR		
6.	a. Explain the for **loop** with syntax and example.	6	L2
	b. What is a **switch statement**? How is it different from an if-else ladder?	6	L2
	c. Write a program to check whether a number is positive, negative, or zero using **if-else**.	8	L3

Course outcomes: Students will be able to

CO1: Described the fundamental programming concepts of C language.

CO2: Explain25 various mechanisms for writing programs using functions.